Оценка:

$0{,}6\,ДЗ + 0{,}4\,ЭКЗ$

$\downarrow$

$0{,}6\,\text{МИН}(ДЗ, 12)$

ЭКЗ УСТНЫЙ

$ДЗ = 4-5$ шт.

$ДЗ = \frac{\sum ДЗ_i}{4/5}$

ЭКЗ:

4 БАЛЛА ТЕОРМИН. 4 мин.

4 ВОПРОСА

4 БАЛЛА до 11 мин

БИЛЕТ

2 БАЛЛА ОСТАТОК времени

3 АГАДКА

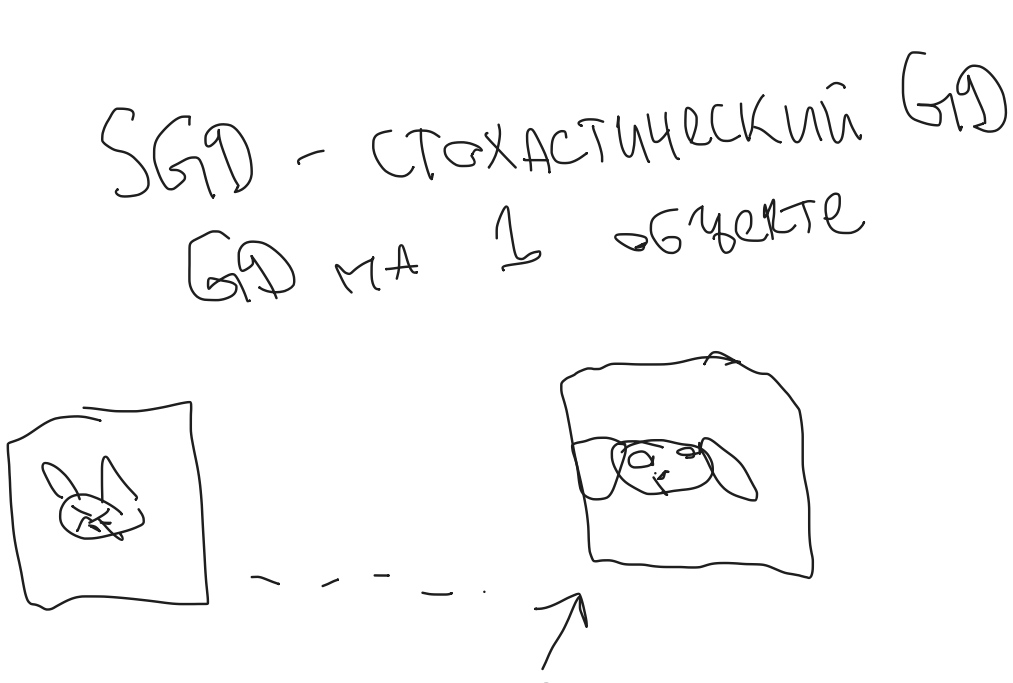
} до 15 минут

GD - град спуск

$L(\cdot,\cdot)$ - ф-ия потерь loss

SGD - стохастический GD

GD на 1 объекте



minibatch SGD

GD на n объектов

$n \geq 2$

B - размер подвыборки (batch_size)

W / θ - обучаемые параметры (Веса, weights)

$f_w(\cdot)$ - нейронка $\{(x_i, y_i)\}_{i=1}^{\ell}$ - датасет

Шаг обучения ∀t:

$$\mathcal{L}^t(w) = \frac{1}{B}\sum_{i=1}^{B} L(f_w(x_i), y_i)$$

$t \sim \mathcal{U}(\{1 \dots k\})$

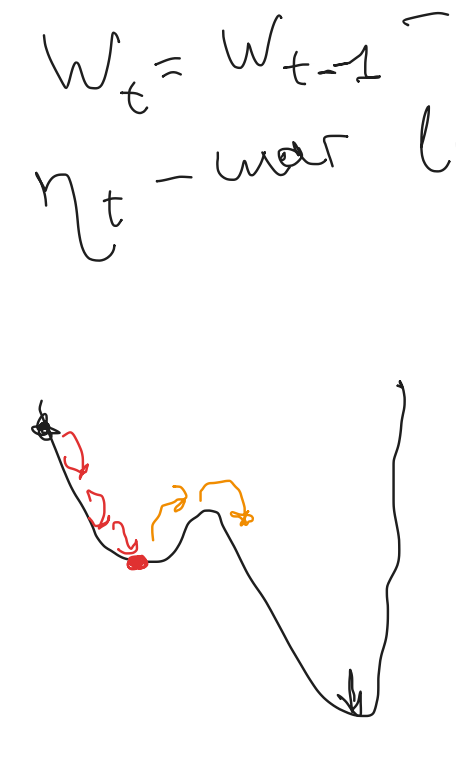
minibatch Везде

$g_t = \nabla_w \mathcal{L}^t(w_{t-1})$

∇ - grad

$w_t = w_{t-1} - \eta_t g_t$

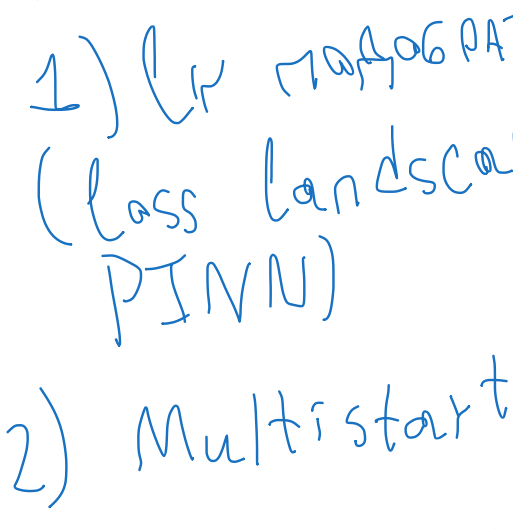
η_t - шаг lr (learning rate)



Решения:

1) lr подобрать (loss landscape PINN)

2) Multistart



Физика:

Инерция

Momentum

$g_t = \nabla \mathcal{L}^t(w_{t-1})$

$m_t = \mu m_{t-1} + g_t \leftarrow$

$w_t = w_{t-1} - \eta_t m_t$

$m_0 = 0 \quad 0 \leq \mu < 1$

$\mu \approx 0{,}9$

Переобучение:

Топ причин:

1) Слишком много параметров

2) . . .

. . .

Лечение:

L1 / L2 регуляризация:

$$\mathcal{L}^t(w) \to \mathcal{L}^t_\lambda(w) := \mathcal{L}^t(w) + \frac{\lambda}{2}\|w\|_2^2$$

$\nabla_w \|w\|_2^2 = 2w$

$g_t = \nabla_w \mathcal{L}^t(w_{t-1}) + \lambda w_{t-1}$

$m_t = \dots$

$$w_t = w_{t-1} - \eta_t m_t = \underline{w_{t-1}} - \eta_t(\mu m_{t-1} + \nabla_w \mathcal{L}^t(w_{t-1}) + \underline{\lambda w_{t-1}}) =$$

$$= w_{t-1}\underbrace{(1-\eta_t\lambda)}_{L2} - \eta_t(\mu m_{t-1} + \nabla_w \mathcal{L}^t(w_{t-1}))$$

λ - weight_decay

Домножение на $\underset{\smile}{<1}$ = L2

Adam (Алгоритм SGD который все берут):

1) Adaptive LR (AdaGrad RMSProp)

2) Momentum

$g_t = \nabla_w \mathcal{L}^t(w_{t-1}) + \lambda w_{t-1}$

$m_t = \beta_1 m_{t-1} + \underline{(1-\beta_1)} g_t \quad \mu \to \beta_1$

$v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2 \to$ element wise

$g_t = (g_1, g_2, g_3)$

$g_t^2 = (g_1^2, g_2^2, g_3^2)$

$$\hat{m}_t = \frac{m_t}{1-\beta_1^t}; \quad \hat{v}_t = \frac{v_t}{1-\beta_2^t}$$

$$w_t = w_{t-1} - \eta_t \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon}$$

$\varepsilon \approx 10^{-6 \dots 8}$ ... (не делим на 0)

$\boxed{m_0 = v_0 = 0}$

$\beta_1 = 0{,}9 \quad \beta_2 = 0{,}999$

$m_t = (1-\beta_1)g_t + \beta_1 m_{t-1} = (1-\beta_1)g_t +$

$+ \beta_1(1-\beta_1)g_{t-1} + \beta_1^2 m_{t-2} =$

$= (1-\beta_1)g_t + \beta_1(1-\beta_1)g_{t-1} + \dots +$

$+ \beta_1^{t-1}(1-\beta_1)g_1 + \beta_1^t \underset{=0}{m_0} =$

$= \sum_{i=0}^{t-1} \beta_1^i (1-\beta_1) g_{t-i}$

$$\hat{m}_t = \frac{m_t}{1-\beta_1^t} = \frac{1-\beta_1}{1-\beta_1^t}\sum_{i=0}^{t-1}\beta_1^i g_{t-i} = \frac{1}{\sum_{j=0}^{t-1}\beta_1^j}\sum_{i=0}^{t-1}\beta_1^i g_{t-i} =$$

$$\frac{1-\beta_1}{(1-\beta_1)(\beta_1^0 + \beta_1^1 + \beta_1^2 + \beta_1^3 \dots)} = \frac{1-\beta_1}{1-\cancel{\beta_1}+\cancel{\beta_1}-\cancel{\beta_1^2}+\cancel{\beta_1^2}\dots} =$$

$$= \frac{1-\beta_1}{1-\beta_1^3 + \dots}$$

$$= \frac{1}{\sum_{j=0}^{t-1}\beta_1^j}\sum_{i=0}^{t-1}\beta_1^i g_{t-i} = \left\{ \alpha_i = \frac{\beta_1^i}{\sum_{j=0}^{t-1}\beta_1^j} : \alpha_i > 0, \ \sum_i \alpha_i = 1 \right.$$

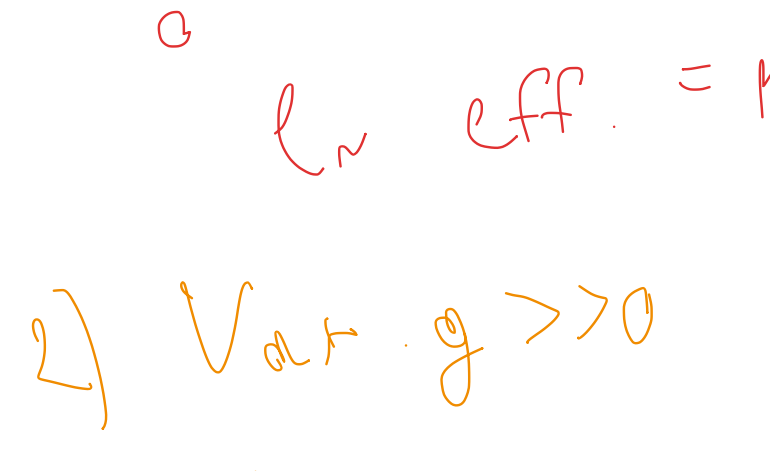
$$\frac{\beta_1^0 + \beta_1^1 + \beta_1^2 + \dots}{\beta_1^0 + \beta_1^1 + \beta_2^2 + \dots} \Bigg\} =$$

$$= \sum_{i=0}^{t-1} \alpha_i g_{t-i} = \mathbb{E} g$$

$\alpha_i > 0$

$\sum \alpha_i = 1$

$$\hat{v}_t = \mathbb{E} g^2 = \mathrm{Var}(g) + (\mathbb{E} g)^2$$



1) $\mathrm{Var}(g) \approx 0$

$$w_t = w_{t-1} - \eta_t \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon}$$

$$\frac{\mathbb{E} g}{\sqrt{\underset{0}{\cancel{\mathrm{Var}(g)}} + (\mathbb{E} g)^2} + \varepsilon} \Rightarrow \approx 1$$

lr eff. $= \eta_t$

2) $\mathrm{Var}\ g \gg 0$

$$\frac{\mathbb{E} g}{\sqrt{\mathrm{Var}(g) + (\mathbb{E} g)^2} + \varepsilon} < 1$$

$\Rightarrow lr < \eta_t$

Adam W:

$$w_t = w_{t-1} - \eta_t \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon} =$$

$$= w_{t-1} - \eta_t \frac{m_t}{(1-\beta_1^t)(\sqrt{\hat{v}_t} + \varepsilon)} = w_{t-1} - \eta_t \frac{\beta_1 m_{t-1} + \nabla_w \mathcal{L}^t(w_{t-1}) + \lambda w_{t-1}}{(1-\beta_1)^t(\sqrt{\hat{v}_t} + \varepsilon)} =$$

$$= w_{t-1}\left(1 - \eta_t \frac{\lambda}{(1-\beta_1^t)\underset{\smile}{(\sqrt{\hat{v}_t} + \varepsilon)}}\right) - \dots$$

Для разных весов разная λ_{eff}

$g_t = \nabla_w \mathcal{L}^t(w_{t-1})$

$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t$

$v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2$

$\hat{m}_t = \frac{m_t}{1-\beta_1^t} \quad \hat{v}_t = \frac{v_t}{1-\beta_2^t}$

$w_t = w_{t-1}(1-\eta_t\lambda) - \eta_t \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon}$

} AdamW

LR scheduler

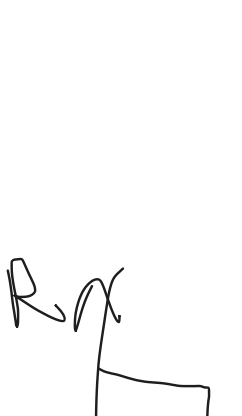
$\eta_t = \text{const}?$

Step LR:

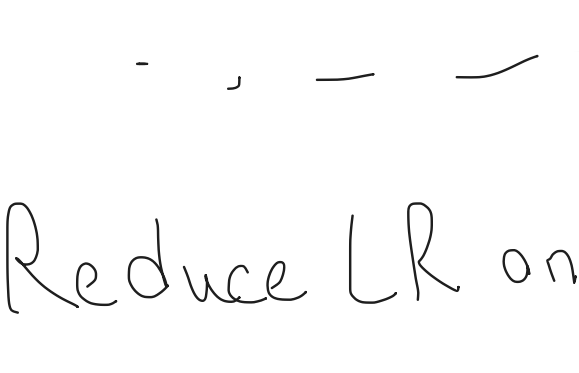
$\eta_0; \quad 1 \leq t \leq t_0$

$\eta_1: \quad t_0 < t \leq t_1$

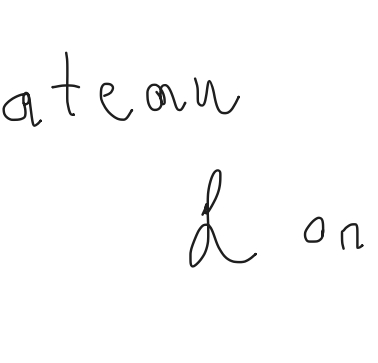
. . . . . .



Reduce LR on Plateau



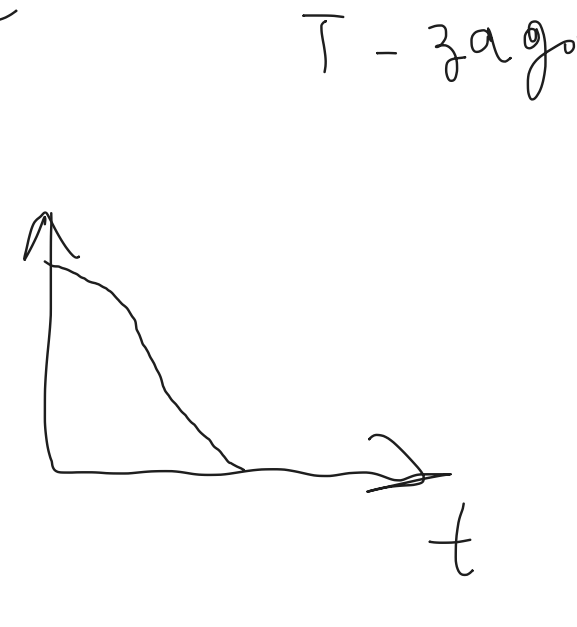
$\mathcal{L}$ on Val



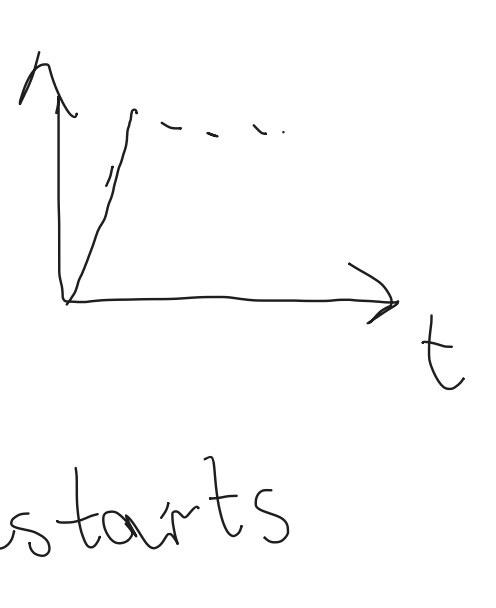
Cosine LR

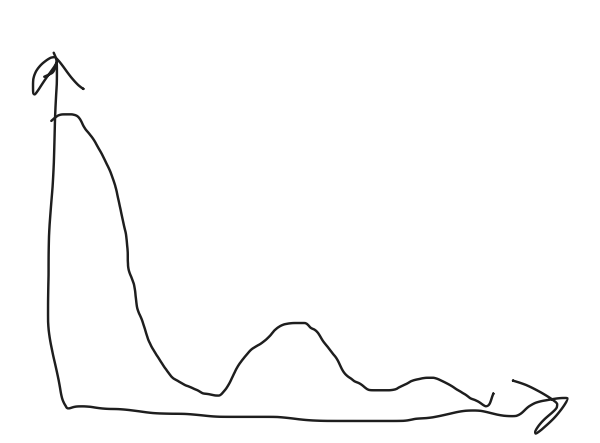
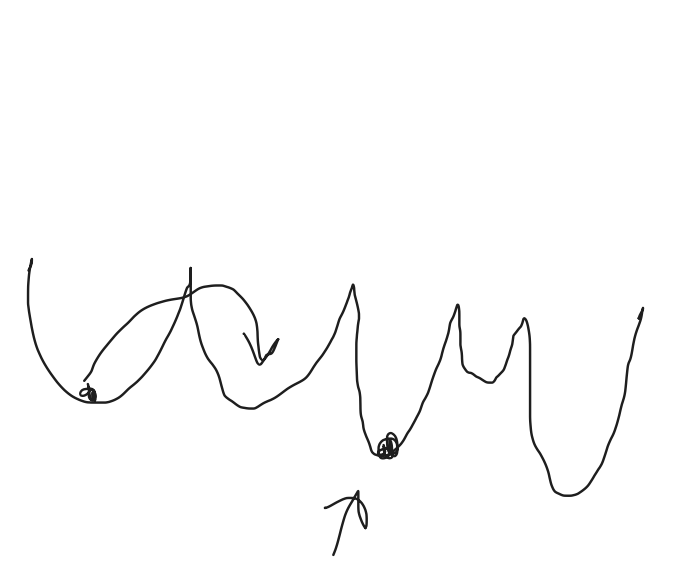
$$\eta_t = \eta_0\left(\frac{1}{2}\left(1 + \cos\left(\frac{\pi t}{T}\right)\right)\right)$$

T - задаем



LR warmup



Restarts